# Career Authenticity Extension

Technical Project Documentation & Architectural Overview

## 1. Project Vision

The Career Authenticity Extension is a high-scale cybersecurity tool designed to protect job seekers from sophisticated recruitment fraud. Unlike basic keyword filters, this system utilizes a **Multi-Layered Verification Pipeline** that analyzes job postings across infrastructure, legal registration, social proof, and linguistic patterns.

## 2. Technological Stack

| Frontend | Backend | Database |
| --- | --- | --- |
| **JavaScript (ES6+)** Chrome Extension API, HTML5/CSS3 (Side Panel UI), DOM Observation. | **Python 3.x** Flask Web Framework, RESTful API architecture, Gunicorn (Production ready). | **MongoDB Atlas** Cloud NoSQL for Global Blacklisting, Audit Logging, and Cache Management. |

## 3. The Logic Pipeline (Working Mechanism)

**Stage 1: DOM Extraction**
The extension parses job metadata (Title, Company, URL, Description) directly from the browser DOM using targeted selectors for LinkedIn, Indeed, and Google Jobs.

**Stage 2: Infrastructure Audit**
Queries **Google Safe Browsing** for phishing reports and **Cloudflare Radar** for global domain ranking to detect "Ghost Sites."

**Stage 3: Legal & Social Identity**
Pings **Sandbox MCA API** for Indian Government registration records and **Apify LinkedIn Actors** to verify human employee footprints.

**Stage 4: Semantic Intelligence (NLP)**
Uses a pre-trained ML model to calculate scam probability based on linguistic anomalies.

## 4. Natural Language Processing (NLP) Layer

The core content analysis utilizes **TF-IDF Vectorization** (Term Frequency-Inverse Document Frequency) to convert text into numerical features. A **Logistic Regression** model then predicts the risk score based on trained weights.

$$P(scam) = 1 / (1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)})$$

- **Heuristic Checks:** Analyzes "Market Logic" (e.g., offered salary vs. average market rate via Adzuna API).
- **Anomaly Detection:** Flags "Too Good To Be True" phrases and urgency-based coercion.

## 5. Key Dependencies

- **Google Safe Browsing:** Real-time threat intelligence.
- **Cloudflare Radar:** Global domain popularity and trust metrics.
- **Sandbox.co.in:** Official MCA (Ministry of Corporate Affairs) Master Data access.
- **Apify:** Distributed web scraping for Social Proof (LinkedIn).
- **Adzuna API:** Live salary benchmarking and market analytics.
- **PyMongo & Certifi:** Secure cloud database integration with MongoDB.

## 6. Solving the "New vs. Legit" Problem

The system employs a **Weighted Scoring Model**. A new website with a low domain age can still achieve a "Safe" score if it possesses a valid Government CIN (Corporate Identification Number) and verifiable social profiles for its founders, effectively filtering out money-laundering shells.

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